The geometric superpotential–FLTZ comparison

Milestone 1: endpoint statement and categorical sufficiency

Working document for a future Lean formalization

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Abstract

We specify a category-free geometric theorem whose conclusion suffices for the stopped comparison in *Fukaya categories of toric superpotentials and the FLTZ stop*. The conclusion records compact Liouville pairs, their witnessing contact forms, a finite chain of smooth deformations and marked completion identifications, and a cotangent ribbon whose full embedded core is exactly the FLTZ link. We prove the conditional passage from these data to the stated categorical result using the general GPS theorems, and audit the hypotheses at that interface.

Status and scope. This milestone fixes the geometric statement and checks its sufficiency. Target Theorem 4.1 is an *unproved target in this document*; the immediate Lean milestone is to define its underlying objects and encode its proposition, before attempting its proof. The lemmas and the conditional implication below have proofs. No Lean code or kernel verification is claimed. In particular, the proposed replacement of the stable-manifold theorem by an elementary trapping estimate has not been proved here. The existing manuscript [4] supplies the comparison to be matched and a proof roadmap, not an implicit proof of the new target. Section 8 specifies the statement and foundation handoff.

1 Input data and the two endpoints

Fix a based lattice $N \simeq \mathbb{Z}^n$, $n \geq 1$, its dual M, and

$$T = M_{\mathbb{R}}/2\pi M, \qquad Y = M_{\mathbb{R}} \times T, \qquad z = \exp(u + i\theta).$$

Let $Q \subset N_{\mathbb{R}}$ be a full-dimensional lattice polytope containing zero in its interior. Set $V = \mathrm{Vert}(Q)$, assume that the vertices of every facet of Q form a lattice basis of N, and let Σ consist of zero and the cones on all nonempty proper faces of Q. Thus Σ is the smooth complete Fano fan determined by these data and V is exactly its set of primitive ray generators. These are the combinatorial inputs of [4]. Fix $c = (c_v)_{v \in V} \in (\mathbb{C}^*)^V$ and no other monomials:

$$W = W_c = \sum_{v \in V} c_v e^{\langle v,u\rangle + i\langle v,\theta\rangle}, \quad \phi_0 = \tfrac{1}{2}|u|^2, \quad \lambda_0 = \sum_j u_j \, d\theta_j.$$

Use $\iota_Z d\lambda = \lambda$ and

$$\lambda_{\mathbb{C}} = \tfrac{1}{2}(x\,dy - y\,dx), \qquad \lambda_K = \lambda_0 + KW^*\lambda_{\mathbb{C}} \quad (K \geq 0).$$

The angle variables are quotient coordinates: the exponentials and the one-forms $d\theta_j$ descend from $M_{\mathbb{R}}$ to T. They do not require globally defined real-valued angles. With the standard complex

structure $J\partial_{u_j} = \partial_{\theta_j}$, W is holomorphic and $d\lambda_K = \sum_j du_j \wedge d\theta_j + KW^*(dx \wedge dy)$ is symplectic: the first summand is positive on (v, Jv) for $v \neq 0$, and the second is nonnegative. Thus Z_K, defined by $\iota_{Z_K} d\lambda_K = \lambda_K$, is well defined for every $K \geq 0$; completeness is a separate conclusion. Completeness means completeness of this Liouville field in both time directions. Finite Liouville type means that inclusion of a compact Liouville domain extends to a primitive-preserving diffeomorphism from its completion onto the given manifold.

Identify $T^*T = T \times N_{\mathbb{R}}$ with primitive $\lambda_{\mathrm{can}} = \sum_j p_j \, d\theta_j$. Choose the Euclidean norm determined by the fixed basis and put

$$D^*T = \{|p| \leq 1\}, \quad S^*T = \{|p| = 1\}, \quad \alpha_{\mathrm{can}} = \lambda_{\mathrm{can}}|_{S^*T}.$$

The target link is the following literal closed subset of S^*T:

$$\begin{aligned} \Lambda_\Sigma^\infty &= \bigcup_{\sigma \in \Sigma,\ \sigma \neq 0} T_\sigma \times (\sigma \cap S^{n-1}), \\ T_\sigma &= \{\theta : \langle v, \theta \rangle \in 2\pi\mathbb{Z} \text{ for every } v \in \sigma \cap N\}. \end{aligned} \tag{1}$$

This specifies the positive-cone, period-2π convention. The FLTZ reference [3, §3.1, equation (8)] provides the formula's name and opposite-cone convention, not a geometric or categorical input. The cotangent lift $(\theta, p) \mapsto (-\theta, -p)$ changes the cone sign. The map $(x, p) \mapsto (2\pi x, p/(2\pi))$ changes period one to period 2π while preserving the cotangent primitive. These maps compare the conic subsets before taking their links in the chosen boundary sections.

2 The geometric data carried by a comparison

Differential-geometric conventions. All manifolds are finite-dimensional, Hausdorff and second countable. Smooth means C^∞, with smooth extension across boundaries and parameter-interval endpoints. A symplectic form is a closed, pointwise nondegenerate two-form. A Liouville domain (D, λ) is a compact smooth manifold with boundary, with $d\lambda$ symplectic and the unique field $\iota_Z d\lambda = \lambda$ strictly outward. Outward means $d\rho(Z) > 0$ for a local defining function $D = \{\rho \leq 0\}$ at each boundary point. In dimension zero these conditions hold for the zero form and zero field on a finite set, whose boundary is empty. A contact form on a $(2n-1)$-manifold means $\alpha \wedge (d\alpha)^{n-1}$ is nowhere zero, including $n = 1$.

Definition 2.1 (Witnessed compact pair)**.** *A pair consists of a compact smooth $2n$-manifold with boundary D, a one-form λ with $d\lambda$ symplectic and Z_λ strictly outward at ∂D, a compact smooth $(2n-2)$-manifold with boundary F, a smooth embedding $j : F \hookrightarrow \partial D$, and a smooth positive function $b : \partial D \to \mathbb{R}_{>0}$, such that*

$$\alpha = \lambda|_{\partial D}, \qquad \mu = j^*(b\alpha)$$

makes F a Liouville domain. All smooth data extend across the indicated boundaries. The completion is $\widehat{D} = D \cup_{\partial D} (\partial D \times [0, \infty))$ with end primitive $e^r\alpha$. Its boundary at infinity is represented by ∂D. When another completed ambient space is named, a marking is a specified primitive-preserving diffeomorphism from $\widehat{D}$ onto that space.

The full intrinsic core and its embedded image are

$$\mathrm{Core}(F, \mu) = \bigcap_{t \geq 0} \mathrm{Fl}_{Z_\mu}^{-t}(F), \qquad j(\mathrm{Core}(F, \mu)) \subset \partial D.$$

The flow is computed in the completion of F. Equivalently the core consists of those $x \in F$ for which $\mathrm{Fl}^t_{Z_\mu}(x) \in F$ for every real $t \geq 0$. Here $\mathrm{Fl}^{-t}(F)$ denotes the image under time $-t$. The Liouville collar supplies the smooth gluing in the completion; the field there is ∂_r. Compactness on the interior and this end model give a flow defined for all real times on the completion. When $n = 1$, a ribbon is a compact zero-dimensional manifold; its Liouville field is zero and its core is the entire ribbon.

Definition 2.2 (Permitted comparison steps)**.** *A* pair deformation *is a smooth family of all the data in Definition 2.1 over $[0, 1]$. It may be given on compact smooth domain and ribbon bundles; a chosen trivialization presents it on fixed D and F. Thus λ_t, j_t, b_t and $\mu_t = j_t^*(b_t \lambda_t|_{\partial D})$ are jointly smooth, and both Liouville fields are outward for every t.*

A completion identification *consists of a specified primitive-preserving diffeomorphism $\Psi : \widehat{D} \to \widehat{D}'$ and its induced contact map $\psi : \partial D \to \partial D'$ at infinity, together with a diffeomorphism $h : F \to F'$ satisfying*

$$\Psi^* \lambda' = \lambda, \qquad \psi \circ j = j' \circ h, \qquad h^* \mu' = c_{\mathrm{scale}} \mu \quad (c_{\mathrm{scale}} > 0 \text{ constant}).$$

The boundary map uses Liouville trajectories to compare the chosen boundary sections; it is part of the data, not an arbitrary identification of abstract cores. A primitive-preserving map intertwines the Liouville fields and hence induces this map at infinity. More explicitly, for $x \in \partial D$, follow the escaping trajectory $\Psi(\mathrm{Fl}^r_Z(x))$ to the unique representative on $\partial D'$; this representative is $\psi(x)$. Equivalently, for large r its end coordinates are $(\psi(x), r + s(x))$ for a smooth real function s. Primitive preservation gives $\psi^* \alpha' = e^{-s} \alpha$.

Write $\mathcal{P} \rightsquigarrow \mathcal{Q}$ for a finite sequence of these steps and their reverses, with an explicit finite list of endpoint tuples. Consecutive steps share their endpoint tuple literally; any change of presentation must be recorded as a permitted step with its maps. If the witnessing forms do not match, an actual pair deformation must be inserted. Neither replacement by a singular subset nor exchange of two ribbons merely because their embedded cores agree is a permitted step. These are deliberate geometric requirements, stronger than the minimum needed for categorical sufficiency.

Remark 2.3. The positive factor b is essential data: arbitrary multiplication of a ribbon primitive by a positive function need not preserve its symplectic differential. No assertion here replaces λ on the ambient manifold by $b\lambda$. A bare exact symplectomorphism with uncontrolled behavior at infinity would also be insufficient for Definition 2.2. Primitive preservation is the stronger condition available for the completion identifications intended here.

Definition 2.4 (Ambient Maslov datum and transport)**.** *For a symplectic manifold (X, ω) choose an ω-compatible almost complex structure J and write $K_J = \Lambda^n_{\mathbb{C}}(T^{1,0}X)^*$ for its complex canonical line bundle. A Maslov datum is a homotopy class of nowhere-vanishing sections of $K_J^{\otimes 2}$. All sections and homotopies in this definition are smooth; section homotopies remain nowhere vanishing and homotopies of complex structures remain compatible with the relevant symplectic form. Compatibility means $\omega(Jv, Jw) = \omega(v, w)$ and $\omega(v, Jv) > 0$ for $v \neq 0$. Changing J through compatible structures identifies these classes; the space of compatible structures is contractible.*

On a pair deformation the datum is carried on the completed ambient family: choose a smooth compatible J_t and a nowhere-vanishing section of the resulting squared canonical line bundle over the whole family. Its restrictions specify the endpoint classes. For a completion identification Ψ, a graded lift is a homotopy from the source section to the pullback of the target section, after identifying the compatible complex structures by a homotopy. A graded chain consists of a chain and these extension and homotopy data, agreeing at junctions. This definition asserts neither existence of a datum on every symplectic manifold nor the prescribed endpoint class for every possible chain.

On Y the initial datum is $[\Omega^{\otimes 2}]$, where $\Omega = \bigwedge_j (du_j + i\,d\theta_j)$ and the standard J above is compatible with every $d\lambda_K$. On T^*T use $J_{\mathrm{can}}\partial_{p_j} = \partial_{\theta_j}$ and the class of $\left(\bigwedge_j (dp_j + i\,d\theta_j)\right)^{\otimes 2}$. Its squared phase on a vertical Lagrangian plane is one; we call this the vertical-phase-zero datum. No grading is assigned to the singular subset Λ_Σ^∞ or to individual branes here.

3 The specified initial ribbon and its admissible choices

The following end bounds specify admissibility without assuming fullness or the desired comparison. Put

$$m_v = c_v e^{\langle v,u\rangle + i\langle v,\theta\rangle}, \qquad G = \sum_v m_v v,$$

$$a = \overline{G}^T G, \qquad b_0 = u \cdot G, \qquad Q_W = |u|^2 a - |b_0|^2 + \tfrac{1}{2}\,\mathrm{Re}(\overline{W} b_0).$$

An ambient end bound $A_{\mathrm{amb}} > 0$ satisfies $a > 0$ and $Q_W > 0$ on $\{\phi_0 \geq A_{\mathrm{amb}}\}$. Write $M(A_{\mathrm{amb}}) = \max_{\{\phi_0 \leq A_{\mathrm{amb}}\}} |W|$; this maximum exists on a nonempty compact set. For $R > \max(1, M(A_{\mathrm{amb}}))$, define the ratio set

$$S_R = \left\{ \frac{|b_0(y)|}{a(y)|W(y)|} : y \in Y,\ |W(y)| \geq R \right\}.$$

Its denominators are positive, since $|W| \geq R$ implies $\phi_0 > A_{\mathrm{amb}}$. When S_R is nonempty and bounded above, put

$$K_0(R) = 1 + 2 \sup S_R. \tag{2}$$

Existence of an ambient end bound, nonemptiness of S_R, and boundedness of S_R are conclusions required of the target. In particular $K_0(R)$ is this specified real number, at least one, rather than an independently chosen threshold. A formalization using a total real supremum must retain the nonemptiness and boundedness assertions explicitly. A fiber end bound $A_{\mathrm{fib}}(R) \geq A_{\mathrm{amb}}$ satisfies

$$a > 0, \qquad |u|^2 - \frac{|b_0|^2}{a} > 0 \quad \text{on } \{\phi_0 \geq A_{\mathrm{fib}}(R),\ |W| \leq R\}. \tag{3}$$

Its existence is likewise part of the target. When $n = 1$, the expression $|u|^2 - |b_0|^2/a$ is zero wherever $a > 0$. Consequently a valid fiber end bound makes the set on which (3) is required empty. Such a bound is possible because $W = c_+ e^{u+i\theta} + c_- e^{-u-i\theta}$ is proper in this case: the triangle inequality shows $|W| \to \infty$ as $u \to \pm\infty$. Do not replace (3) by a positivity claim on a nonempty end in dimension one.

For such bounds and a fixed $K \geq K_0(R)$ choose

$$A_- > A_{\mathrm{fib}}(R), \qquad A > A_-.$$

Use a smooth convex function $m_\varepsilon : \mathbb{R}^2 \to \mathbb{R}$ with nonnegative partial derivatives whose sum is one, satisfying

$$\max(x,y) \leq m_\varepsilon(x,y) \leq \max(x,y) + \varepsilon, \qquad m_\varepsilon(x,y) = \max(x,y) \text{ if } |x-y| \geq \varepsilon.$$

Define

$$\begin{aligned} E_\varepsilon &= \{m_\varepsilon(\phi_0 - A,\ |W|^2 - R^2) \leq 0\}, \\ F_R^- &= \{W = R,\ \phi_0 \leq A_-\}, \qquad \mu_R = \lambda_0|_{F_R^-}. \end{aligned} \tag{4}$$

Definition 3.1 (Admissible rounding)**.** *For these fixed R, K, A_-, A, a rounding consists of a real ε and a profile m_ε satisfying all the smoothness, convexity, derivative and equality conditions above. It is admissible when $0 < \varepsilon < A - A_-$ and, at every point of the explicitly defined band*

$$B_\varepsilon = \{-\varepsilon < \phi_0 - A \le 0, \quad -\varepsilon < |W|^2 - R^2 \le 0\},$$

one has

$$d(|W|^2)(Z_K) > 0, \qquad d\phi_0 \wedge d(|W|^2) \neq 0.$$

The second condition means linear independence of the two real covectors. It is allowed to hold vacuously when B_ε is empty. These conditions do not assume a completion marking, core equality, or comparison chain. Existence of admissible roundings and the asserted regularity of their domains are conclusions of the target.

For orientation, set $E_0 = \{\phi_0 \le A,\ |W| \le R\}$. The profile conditions imply $E_\varepsilon \subset E_0$ and $E_0 \setminus E_\varepsilon \subset B_\varepsilon$. Indeed $m_\varepsilon(x+t, y+t) = m_\varepsilon(x,y) + t$, by the sum of the partial derivatives. If $x \le -\varepsilon$ and $y \le 0$, monotonicity gives $m_\varepsilon(x,y) \le m_\varepsilon(-\varepsilon, 0) = 0$; interchange x, y for the other case. The portion of the zero level on which $m_\varepsilon >$ max also lies in B_ε. To see the strict band bounds there, use $m_\varepsilon(x,y) = x + g(y-x)$, where g is nondecreasing, $g(\varepsilon) = \varepsilon$, and $g'(\varepsilon) = 1$. Hence $g(s) < \varepsilon$ for $s < \varepsilon$, whereas $g(s) = s$ for $s \ge \varepsilon$. If $x \le -\varepsilon$ and $y < 0$, these two cases both give $m_\varepsilon(x,y) < 0$; the other case is symmetric. Unrounded faces need not lie in B_ε. The band is disjoint from F_R^- because $\phi_0 > A - \varepsilon > A_-$ there. For $n = 1$ it is empty, $E_\varepsilon = \{|W| \le R\}$, and $F_R^- = W^{-1}(R)$. On the unrounded face containing the ribbon, its inclusion j_R has

$$j_R^*(\lambda_K|_{\partial E_\varepsilon}) = \mu_R,$$

since W is constant on F_R^-. The initial witnessing factor is one.

4 The geometric endpoint theorem to be proved

Target theorem 4.1 (Geometric endpoint with specified bounds and grading)**.** *For every based lattice, polytope and coefficient tuple of Section 1, the following assertions hold with the nested quantifiers stated below.*

(G1) *For every real $K \ge 0$, (Y, λ_K) is complete and of finite Liouville type. There is an ambient end bound A_{amb} as in Section 3.*

(G2) *The ambient bound in (G1) can be chosen together with $R_0 > \max(1, M(A_{\mathrm{amb}}))$ so that, for every real $R \ge R_0$, S_R is nonempty and bounded above, and at least one fiber end bound $A_{\mathrm{fib}} \ge A_{\mathrm{amb}}$ exists. For every such R, for every valid fiber end bound (relative to this A_{amb}), for every real $K \ge K_0(R)$, and for every pair of levels $A_- > A_{\mathrm{fib}}$ and $A > A_-$, there exists an admissible rounding as specified in Section 3. For every such choice of levels and rounding, (4) defines a witnessed pair with nonempty ribbon. In particular R is a regular value of W and the full fiber is smooth. Inclusion extends to specified primitive-preserving diffeomorphisms*

$$(\widehat{E_\varepsilon, \lambda_K}) \longrightarrow (Y, \lambda_K), \qquad (\widehat{F_R^-, \mu_R}) \longrightarrow (W^{-1}(R), \lambda_0|_{W^{-1}(R)}).$$

These markings are induced by the corresponding Liouville flows and the given inclusions. Thus the ribbon is a truncation of the whole fiber. The fiber marking preserves the source theorem's fullness assertion; the categorical deduction itself only uses the ambient marking.

(G3) *For each of those initial pairs there exist a compact smooth ribbon* F_Σ*, an embedding* $j_\Sigma : F_\Sigma \hookrightarrow S^*T$*, and a smooth positive* $b_\Sigma : S^*T \to \mathbb{R}_{>0}$ *such that*

$$\mu_\Sigma = j_\Sigma^*(b_\Sigma \alpha_{\mathrm{can}})$$

is Liouville, and there is a finite chain

$$(E_\varepsilon, \lambda_K, F_R^-, j_R, 1) \rightsquigarrow (D^*T, \lambda_{\mathrm{can}}, F_\Sigma, j_\Sigma, b_\Sigma)$$

in the sense of Definition 2.2. The endpoint data and this chain are chosen together with the grading data specified below. The source marking is the one in (G2); the terminal marking is the standard completion of D^*T *with* T^*T*. The endpoint satisfies the equality of embedded subsets*

$$j_\Sigma\big(\mathrm{Core}(F_\Sigma, \mu_\Sigma)\big) = \Lambda_\Sigma^\infty \subset S^*T. \tag{5}$$

The same chain *is equipped with ambient Maslov data and graded lifts in the sense of Definition 2.4. At its source the datum is the pullback of* $[\Omega^{\otimes 2}]$ *by the marking in (G2). At its endpoint, under the standard cotangent marking, the transported datum is homotopic to the vertical-phase-zero datum of Definition 2.4. This is an existential requirement on the chosen chain, not a claim about every chain between these endpoints.*

Quantifier order. For each input, all-K completeness and finite type are asserted first. There exist A_{amb} and R_0 satisfying the stated conditions. For every $R \geq R_0$, assert nonemptiness and boundedness of S_R and existence of a fiber end bound; then define $K_0(R)$ by (2). For *every* valid fiber end bound, every $K \geq K_0(R)$, and every $A_- > A_{\mathrm{fib}}$, $A > A_-$, assert existence of an admissible rounding. For *every* admissible rounding, assert the witnessed pair and both specified markings, followed by one existential package containing the endpoint ribbon, witness, chain, embedded-core equality, and grading extensions and endpoint homotopy. The existence clauses precede the universal admissibility clauses and are not replaced by them. The endpoint ribbon and the chain need not be independent of R, K or the choices; their embedded core is always the fixed set (1). Item (G1) is asserted also below the ribbon threshold. The target assumes neither a strongly compatible potential, a full-core identification, nor the desired chain as input. Their construction belongs to its eventual proof.

Lemma 4.2 (Comparison of initial choices)**.** *Assume (G1)–(G2). Fix the input and* R, K *in their stated range. Any two admissible initial pairs, including choices using different valid fiber bounds, are joined by a finite chain inside the fixed* (Y, λ_K)*. Domain primitives are restrictions of* λ_K*, the completion markings extend inclusion, and every completion identification is the identity on the marked* Y*. The chain preserves* $[\Omega^{\otimes 2}]$ *at both ends and transports any fixed ambient background class identically.*

Proof. Let the two fiber levels be A_-^0, A_-^1. The smaller of their valid fiber end bounds is itself a valid bound, and both levels exceed it. Choose $L > \max(A_-^0, A_-^1)$ above this bound, then $A'' > L$ and an admissible rounding for these levels. Its domain E'' has $A'' - \varepsilon'' > L$, so both ribbons, and every intermediate fiber sublevel, lie in its unrounded $|W| = R$ face.

For each endpoint level, identify its original ambient completion with $\widehat{E}''$ using the two inclusion markings. The boundary map fixes the ribbon pointwise, because that ribbon lies on both boundary sections. Its primitive is μ_R on both sides, so take $h = \mathrm{id}$ and $c_{\mathrm{scale}} = 1$. Inside E'' vary the ribbon level linearly from A_-^0 to A_-^1. On a regular holomorphic fiber, the intrinsic Liouville field is the orthogonal projection of $Z_0 = \sum_j u_j \partial_{u_j}$ onto the fiber tangent space. Therefore its derivative of

ϕ_0 is $|u|^2 - |b_0|^2/a$. The fiber-end inequality makes $d\phi_0$ nonzero on that tangent space throughout this compact band and makes the field outward. A transverse vector field on a neighborhood of the band, cut off outside it, gives smooth identifications of these sublevel domains. Thus this is a pair deformation, with ambient primitive λ_K and witness one. The two completion identifications and this deformation give the required finite chain. If ribbon presentations change in the chosen trivialization, record the resulting relabeling as an identity ambient completion step. Pull back the fixed ambient datum and any background class along the inclusion markings throughout. □

Brane conventions. For the categorical consequence fix a field and the Spin conventions of GPS1 [1, §3], with zero ambient background class. Modulo-two gradings use the underlying geometric chain; integer gradings use the additional data in (G3). Coefficients and integer gradings use the setup of GPS2 [2, §2.2]. If relative-spin conventions with a nonzero background class are additionally adopted, their categorical foundations are a separate input. Such a class is transported along the chosen chain; no independence of that endpoint class from the chain is asserted. Lemma 4.2 does preserve fixed ambient background data for comparisons of initial choices. There is no assertion about the shifts, spin structures, or identification of named objects.

5 Why these geometric data suffice

The geometric construction in the next lemma precedes any categorical statement. We use the general pair-to-sector construction of GPS1 [1, Definition 2.14, Lemma 2.13, Remark 2.12, and Lemma 2.32].

Lemma 5.1 (Realizing a compact pair family by sectors)**.** *Every pair deformation in Definition 2.2 admits a smooth family of associated Liouville sectors whose convex completions identify with the original completed ambients and their embedded boundary ribbons. Any specified ambient markings are preserved by these identifications.*

Proof. Trivialize the compact bundles. Write $\alpha_t = \lambda_t|_{\partial D}$. Choose one constant C so $C + \log b_t(x) > 0$ for every (t, x). The outward graphs $r = C + \log b_t(x)$ in the completed ends bound Liouville domains D'_t. Their boundary forms, pulled back to ∂D, are $e^C b_t \alpha_t$. The lifted ribbons have primitive $e^C \mu_t$; this constant factor leaves their Liouville fields and cores unchanged. The completions identify by inclusion with the original completions. The lifted ribbons and the original ribbons represent the same subsets at infinity, since corresponding points lie on the same Liouville rays.

Reeb flow gives, uniformly over the compact parameter interval, contact collars with form $dz + e^C \mu_t$. On this collar the vector field $V_t = z\partial_z + Z_{\mu_t}$ satisfies $\mathcal{L}_{V_t}(dz + e^C \mu_t) = dz + e^C \mu_t$. It is strictly outward on each face of $F \times [-\delta, \delta]$. Its contact Hamiltonian is z, whose Reeb derivative is one. These are exactly the conditions $\mathbf{R}_\lambda I > 0$ and V_I outward along ∂A in GPS1 Lemma 2.13, with $I = z$ and $A = F \times [-\delta, \delta]$. Compactness makes the strict inequalities uniform, so collars and corner roundings can be chosen smoothly in t. The linear homogeneous extension of $-z$ gives the defining function on the resulting sector boundary, as in the cited construction; its positivity persists under the small roundings. The resulting sectors vary smoothly; the cited completion construction recovers the marked completed ambients. This argument applies also when F is zero-dimensional. □

Sector deformations here allow the structure at infinity to vary, as in GPS1 [1, Lemma 2.9 and the proof of Corollary 4.38]; they are not restricted to compactly supported variations. GPS2 [2, §1.1] states invariance directly for Liouville pairs and gives the pair-language ribbon and core

equivalences. The categorical inputs are now isolated: GPS2 [2, Lemma 3.4] gives invariance under sector deformation, and [2, Corollary 3.9] compares a sector with its completed ambient stopped at the ribbon or at its full core. Neither input here requires a Weinstein ribbon or a mostly Legendrian stop. Their Floer foundations remain external to this document.

Proposition 5.2 (Categorical sufficiency of the target)**.** *Assume Target Theorem 4.1. Under the preceding GPS inputs and brane conventions, for every stated R, K and admissible initial ribbon,*

$$\operatorname{Perf} \mathcal{W}(Y, \lambda_K; F_R^-) \simeq \operatorname{Perf} \mathcal{W}(T^*T, \lambda_{\mathrm{can}}; \Lambda_\Sigma^\infty). \tag{6}$$

The left-hand equivalence type is independent of the admissible truncations and roundings. The integer-graded conclusion has the specified endpoint Maslov datum.

Proof. Item (G2) identifies the source with the stopped completion of the initial witnessed pair, with its actual embedded ribbon. For each deformation step in (G3), Lemma 5.1 provides a sector family. The two categorical inputs identify the categories at its endpoints stopped at their ribbons. A completion-identification step gives the same conclusion by transporting the ambient and ribbon through its specified primitive-preserving map (transport of the Floer setup as in GPS2 [2, §2.5]). Compose these comparisons along the finite chain. At this stage the target stop is the smooth ribbon $j_\Sigma(F_\Sigma)$. Apply the ribbon/core comparison there, using the witnessing primitive μ_Σ. Equation (5) identifies the resulting stop with the literal FLTZ link in the specified cotangent boundary. This proves the comparison before taking perfect envelopes; taking idempotent-complete pretriangulated envelopes preserves it.

Independence of equivalence type already follows because each admissible choice has been compared with the same cotangent stopped category. Lemma 4.2 additionally supplies comparisons preserving the fixed ambient data. For integer gradings, transport the data on the chosen chain in (G3) and use its endpoint homotopy. Ribbon/core replacement changes no ambient grading datum. Transport the remaining brane conventions as specified above. □

Remark 5.3 (Exact categorical scope). Under the declared categorical foundations, Equation (6) recovers the stopped-model comparison of [4, Theorem 1.1], with its parameter range, admissible-choice assertion and ambient grading convention. Here the admissible levels are explicitly allowed for every valid fiber end bound relative to the supplied ambient bound. The notation FS refers to this stopped Landau–Ginzburg model. No Morse condition on W is assumed. A comparison with a separately defined directed or thimble category, or identification of distinguished objects, needs further hypotheses and arguments; it is not a consequence of Proposition 5.2. The geometric target is sufficient for the categorical result, not asserted to be logically necessary for all possible proofs of that result.

6 The new geometric junction: a feasibility check

The existing manuscript uses a categorical ribbon/core replacement at an intermediate junction. The following elementary lemma explains why a finite geometric deformation is available for the particular nested ribbons planned here. Its hypotheses remain obligations of the toric proof.

Lemma 6.1 (Finite intrinsic shrinking)**.** *Let a complete exact symplectic manifold (H, μ_H) be the marked completion of a compact Liouville domain $F_0 \subset H$. Suppose F_0 lies as a ribbon in an unaltered smooth face of a contact boundary, with restricted contact form $\mu_H|_{F_0}$. For every open neighborhood U of $\operatorname{Core}(F_0, \mu_H|_{F_0})$ in H, there is a finite $T \geq 0$ such that $F_T = \operatorname{Fl}_{Z_H}^{-T}(F_0) \subset U$. For $0 \leq \tau \leq T$, these domains form a smooth ribbon deformation inside that same face. With F_0 as source, the primitive is $e^{-\tau}\mu_H|_{F_0}$.*

Proof. Outwardness along ∂F_0 implies negative invariance of F_0. The domains $F_\tau = \mathrm{Fl}_{Z_H}^{-\tau}(F_0)$ are therefore nested compact domains in F_0. Their intersection is the core by definition. If no F_m for an integer $m \geq 0$ lay in U, the nonempty compact sets $F_m \setminus U$ would be nested in F_0, giving a point in the core outside U, a contradiction. Completeness and smooth flow give the embeddings throughout the finite interval. Cartan's formula and $\mu_H(Z_H) = 0$ give $\mathcal{L}_{Z_H}\mu_H = \mu_H$, hence the asserted pullback identity. Multiplication by a positive constant preserves the Liouville field and its outwardness. The domains remain in F_0, hence in the given face. □

Conditional application and notation. The following notation describes the planned application from [4, §§4 and 7], rather than supplying its missing hypotheses. Write f_1 for the final normalized tailored function, λ_ϕ for the adapted ambient primitive, and $\lambda_{1,K} = \lambda_\phi + K f_1^* \lambda_{\mathbb{C}}$. Let $H_\beta = f_1^{-1}(1)$ with intrinsic primitive $\mu_H = \lambda_\phi|_{H_\beta}$. At $K = K_*$, E_1 denotes the rounded initial domain for these tailored data and F_1^- its full ribbon in the unrounded face. Let D_β be the larger compact Liouville domain used for the hitting construction, and let $\pi_K(x) = \mathrm{Fl}_{Z_{1,K}}^{t_K(x)}(x)$ be the map to its boundary. These domains, smooth hitting times, and their required properties remain to be constructed in a proof of the target. In particular, the proposed junction requires inclusion markings $q_E : \widehat{E}_1 \to (\bar{X}, \bar{\lambda})$ and $q_D : \widehat{D}_\beta \to (\bar{X}, \bar{\lambda})$ onto the same completed ambient, agreeing with the inclusion $E_1 \subset D_\beta$ on the compact pieces. One may take $\bar{X} = \widehat{(D_\beta, \lambda_{1,K_*})}$ and $q_D = \mathrm{id}$. The intended strict completion map is $q_D^{-1} \circ q_E$; nesting alone does not assert these markings. This requires no global completion identification with the tailored (Y, λ_{1,K_*}): the planned ambient estimates are on the compact domain and its finite band, while whole-fiber completeness is a separate intrinsic assertion.

In this application $F_0 = F_1^-$ in the tailored hypersurface. The common neighborhood U is one on which all the ambient hitting maps π_K, $0 \leq K \leq K_*$, are embeddings. Establishing that neighborhood and the full-core equality is part of the later proof. Set $F = F_T$ after applying Lemma 6.1. The endpoint presentation on F_0, with embedding $\mathrm{incl} \circ \mathrm{Fl}^{-T}$ and primitive $e^{-T}\mu_H|_{F_0}$, is relabeled to the domain F_T by a completion-identification step with ambient map id, ribbon map $\mathrm{Fl}^{-T} : F_0 \to F_T$, and $c_{\mathrm{scale}} = 1$. The prescribed boundary identification $\psi : \partial E_1 \to \partial D_\beta$ uses the same ambient trajectories as π_{K_*}, so it must satisfy $\psi|_F = \pi_{K_*}|_F$, not merely agreement on the core. The required justification is that E_1 lies inside the negatively invariant D_β and each escaping trajectory crosses its boundary exactly once. This nesting and crossing assertion is another geometric obligation of the proposed application.

Here is the required contact-form matching at this junction. Here α_E and α_D are the restrictions of λ_{1,K_*} to the respective boundaries. If $\psi^*\alpha_D = e^s \alpha_E$, define $b_{\mathrm{in}} = e^{-s} \circ \psi^{-1}$. The hitting-time identity gives $s|_F = t_{K_*}$ and $\pi_{K_*}^* \alpha_D = e^{t_{K_*}} \mu_H|_F$. Consequently $b_{\mathrm{in}} \circ \pi_{K_*} = e^{-t_{K_*}}$. The witness used by the hitting-map family can be chosen with this same pointwise restriction. If its extension to the rest of the boundary is b_*, interpolate by

$$b_\tau = \exp\big((1-\tau)\log b_{\mathrm{in}} + \tau \log b_*\big), \qquad 0 \leq \tau \leq 1.$$

Its restriction to the ribbon is constant in τ, so the pulled-back primitive stays exactly $\mu_H|_F$. This is a permitted pair deformation. Lemma 5.1's radial graphs realize the witnessing forms as actual compact boundary forms up to one constant. Thus no categorical core replacement is needed at this junction, provided the listed geometric hitting-map assertions are proved.

7 Sufficiency audit and remaining proof obligations

Requirement of the original result	Explicit geometric provision	Consequence checked here
All finite $K \geq 0$	(G1), separately from the large-K ribbon construction	Completeness and finite type are not weakened to the comparison range.
Every large R and every $K \geq K_0(R)$	(2) and (G2)–(G3)	No single choice of K replaces the stated conclusion.
Specified full far-fiber stop	(4), both markings in (G2)	The source is the stated embedded ribbon, not an arbitrary copy of its fiber.
Varying ambient primitives	Definition 2.2 and Lemma 5.1	An actual sector family is supplied before deformation invariance is used.
Valid terminal ribbon	$b_\Sigma > 0$ and $\mu_\Sigma = j_\Sigma^*(b_\Sigma \alpha_{\mathrm{can}})$ Liouville	The primitive used to define the full core is specified.
Correct target stop	Equality (5) in S^*T	The endpoint core replacement gives the precise FLTZ stop.
Admissible-choice independence	Lemma 4.2, including its fixed ambient grading	The comparison of initial choices is supplied, not inferred from one example.
Integer gradings	The grading data on the same chain in (G3)	The endpoint datum is identified without grading the singular link or fixing named-object shifts.
No additional categorical theorem	Proposition 5.2 and Remark 5.3	The deduction uses general GPS statements, not a toric application, CCC, HMS, or a thimble-generation assertion.

Proof obligations for the next milestone. The geometric target must be proved from the finite input in Section 1. In dependency order, this requires construction of the compatible convex potential; the end bounds and completeness; the original marked pair and compact tailoring families; all critical points and the strict intrinsic Lyapunov function; the full embedded core; smooth parameter-dependent hitting maps and a common ribbon neighborhood; the finite chain and endpoint core equality; and the ambient grading transport. In particular, neither (5) nor chain existence may be hidden in an admissible-potential hypothesis.

The intended parameter order is: fix coefficient and potential paths and analytic constants; choose inner and outer radius constants; choose $\beta = \log R$ sufficiently large; then choose finite normalized K_*. For the fixed coefficient/potential path (f_t, ϕ_t), first establish the finite path-uniform threshold $\kappa_{\mathrm{path}}(R) = \sup_t \kappa(f_t, \phi_t; R)$ of [4, §4], including $\kappa_{\mathrm{path}}(R) \geq K_0(R)$, and choose $K_*/R^2 \geq \kappa_{\mathrm{path}}(R)$. First connect the supplied original K to K_*/R^2 while keeping the original data fixed; this interval remains above $K_0(R)$. Then traverse the coefficient/potential path at that coefficient. Quadratic homogeneity gives $K_*(f/R)^*\lambda_{\mathbb{C}} = (K_*/R^2) f^*\lambda_{\mathbb{C}}$ for any f; apply this at the path endpoint polynomial when passing to normalized tailoring data. Only afterward choose collars, hitting neighborhoods, and shrinking times. These later choices may depend on the fixed

compact parameter intervals. No estimate for the growth of $K_*(R)$ is part of the target.

8 Handoff for the Lean statement and its foundations

The first formalization task is to define the objects above and encode Target Theorem 4.1 as a proposition. It is not to supply a placeholder proof. A named definition of that proposition can record the unproved target without introducing an axiom or a theorem with a missing proof. Auxiliary structures may bundle geometric objects and proofs of their defining properties; the existence of the target's objects remains the conclusion, not a field of the input data.

Foundations in dependency order.

1. Encode the based finite-rank lattice, its real dual spaces, the finite polytope and face conditions, nonzero coefficients, the quotient torus and its descended forms. Define the fan, W, λ_K and the literal subset (1). A coordinate model is acceptable provided its periodicity and descent to these quotients are proved.

2. Define differential forms, symplectic and contact nondegeneracy, smooth manifolds with boundary, Liouville fields and outwardness, smooth flows and completions. Include dimension zero. Define the core by the all-forward-time condition and connect it to the nested-image formula in Definition 2.1.

3. Define witnessed pairs, inclusion markings, compact parameter families, strict completion identifications and their induced boundary maps. A finite chain records its intermediate pairs and steps with matching endpoints; it is not an arbitrary relation supplied as input. The definition of smooth family must cover the completions as well as the compact pieces used to build them.

4. Define compatible almost complex structures, squared canonical line bundles, homotopies of nonvanishing sections, and the transport data of Definition 2.4. The source and cotangent endpoint classes are the explicit ones specified there.

5. Define the end-bound and rounding predicates, S_R, its nonemptiness and boundedness requirements, and the prescribed $K_0(R)$. Assemble the target with exactly the quantifier order above, including both markings and one chain carrying both core equality and grading. Record Lemma 4.2 as a separate conditional consequence.

Bridges that must be proved or explicitly left pending. The foundation layer needs the well-definedness of quotient formulas; the symplectic calculation for $d\lambda_K$; the Liouville collar and complete flow on a completion; flow equivariance and the boundary map of a strict identification; and the correspondence between geometric families and any chosen coordinate encoding. An orbit characterization of a completion marking requires a proof that it produces the specified smooth primitive-preserving diffeomorphism. It cannot replace the marking by an unrelated predicate. Homotopy invariance under a change of compatible complex structure likewise needs a foundation lemma.

The general existence of the objects asserted in (G1)–(G3) is the later geometric theorem. In particular, end bounds, boundedness of S_R, admissible roundings, full-fiber markings, the full embedded core, the finite comparison chain, and its endpoint grading must not be hidden in an input assumption. A modular partial development should state precisely which foundational bridges and geometric results remain pending. The paper's GPS deduction is an external interface; the first Lean statement does not need a formal Fukaya category.

Checks before accepting the statement. Check the $n = 1$ zero-dimensional ribbon and empty corner band, the positive cone and period-2π conventions, the two time directions of completeness, and the formula-defined threshold for every $R \geq R_0$. Check that changing the fiber bound does not change the quantifier meaning and that the graded chain is exactly the chain furnishing (5). Pin the Lean and Mathlib versions before choosing APIs. Verify that the smoothness notation means C^∞ in that version, rather than analytic regularity, and inspect supremum conventions instead of relying on default values for an empty or unbounded set. No claim about current library coverage is made here. Compilation of definitions checks their types; it does not prove the existence theorem or the fidelity of an abstraction.

Review record. See Claude's prior sufficiency review and the revision and scoped review record. These records concern the specification and the supplied conditional arguments. The geometric existence theorem remains unproved here.

References

[1] S. Ganatra, J. Pardon, and V. Shende, *Covariantly functorial wrapped Floer theory on Liouville sectors*, arXiv:1706.03152v3. https://arxiv.org/html/1706.03152v3.

[2] S. Ganatra, J. Pardon, and V. Shende, *Sectorial descent for wrapped Fukaya categories*, arXiv:1809.03427v4. https://arxiv.org/html/1809.03427v4.

[3] B. Fang, C.-C. M. Liu, D. Treumann, and E. Zaslow, *T-duality and homological mirror symmetry of toric varieties*, arXiv:0811.1228v4. https://arxiv.org/html/0811.1228v4.

[4] *Fukaya categories of toric superpotentials and the FLTZ stop: An explicit far-fiber comparison for smooth Fano fans*, working manuscript, 2 October 2026. Local source: `../research/fukaya-superpotential-fltz.tex`. Theorem 1.1 is the comparison whose scope is audited here.